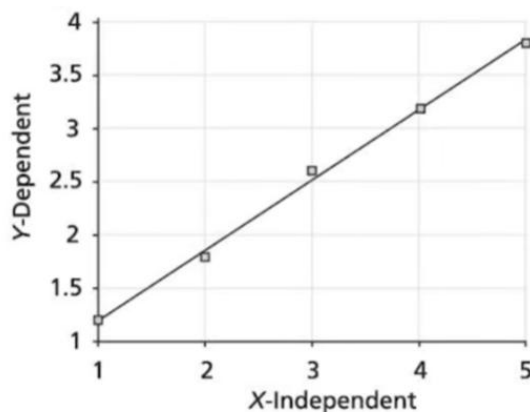


Linear Regression – Solved Example

- Let us consider an example where the five weeks' sales data (in Thousands) is given as shown in Table.
- Apply linear regression technique to predict the 7th and 12th week sales.

x_i (Week)	y_j (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Linear Regression – Solved Example



x_i (Week)	y_j (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Linear Regression – Solved Example

- Linear regression equation is given by

- $y = a_0 + a_1 * x + e$

- where

- $a_1 = \frac{(\overline{xy}) - (\bar{x})(\bar{y})}{\overline{x^2} - \bar{x}^2}$

- $a_0 = \bar{y} - a_1 * \bar{x}$

x_i (Week)	y_j (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Linear Regression – Solved Example

- Here, there are 5 items, i.e., $i = 1, 2, 3, 4, 5$.

	x_i (Week)	y_j (Sales in Thousands)	x_i^2	$x_i * y_j$
	1	1.2	1	1.2
	2	1.8	4	3.6
	3	2.6	9	7.8
	4	3.2	16	12.8
	5	3.8	25	19
Sum	15	12.6	55	44.4
Average	$\bar{x} = 3$	$\bar{y} = 2.52$	$\overline{x^2} = 11$	$\overline{xy} = 8.88$

Linear Regression – Solved Example

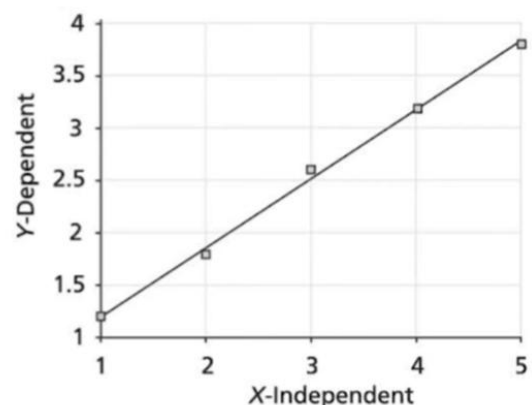
- $\bar{x} = 3$ $\bar{y} = 2.52$ $\overline{x^2} = 11$ $\overline{xy} = 8.88$

- $a_1 = \frac{(\overline{xy}) - (\bar{x})(\bar{y})}{\overline{x^2} - \bar{x}^2} = \frac{8.88 - 3 \cdot 2.52}{11 - 3^2} = 0.66$
- $a_0 = \bar{y} - a_1 * \bar{x} = 2.52 - 0.66 * 3 = 0.54$
- Regression equation is
- $y = a_0 + a_1 * x$
- $y = 0.54 + 0.66 * x$

Linear Regression – Solved Example

- $\bar{x} = 3$ $\bar{y} = 2.52$ $\overline{x^2} = 11$ $\overline{xy} = 8.88$

- $a_1 = \frac{(\overline{xy}) - (\bar{x})(\bar{y})}{\overline{x^2} - \bar{x}^2} = \frac{8.88 - 3 \cdot 2.52}{11 - 3^2} = 0.66$
- $a_0 = \bar{y} - a_1 * \bar{x} = 2.52 - 0.66 * 3 = 0.54$
- Regression equation is
- $y = a_0 + a_1 * x$
- $y = 0.54 + 0.66 * x$



Linear Regression – Solved Example

- Regression equation is
- $y = a_0 + a_1 * x$
- $y = 0.54 + 0.66 * x$
- The predicted 7th week sale (when $x = 7$) is,
- $y = 0.54 + 0.66 * 7 = 5.16$
- the predicted 12th week sale (when $x = 12$) is,
- $y = 0.54 + 0.66 * 12 = 8.46$